



NORTHERN BEACHES SECONDARY COLLEGE

MANLY SELECTIVE CAMPUS

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

2018

Mathematics Extension II

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Write your Student Number at the top of each page
- Answer Section I- Multiple Choice on Answer Sheet provided
- Answer Section II – Free Response in a separate booklet for each question.
- Answer Section II - Question 13d in the supplied additional sheet
- NESA approved calculators and templates may be used.

Section I Multiple Choice

- 10 marks
- Attempt all questions
- Allow about 15 minutes for this section

Section II – Free Response

- 90 marks
- Each question is of equal value
- All necessary working should be shown in every question.
- Allow about 2 hours 45 minutes for this section

Weighting: 40%

Section 1: Multiple Choice (10 marks)

Indicate your answer on answer sheet provided.

Allow approximately 15 minutes for this section.

Q1. A hyperbola has the equation $x^2 - 4y^2 = 4$. The distance between its two directrices is:

A $\sqrt{5}$

B $\frac{4\sqrt{5}}{5}$

C $2\sqrt{5}$

D $\frac{8\sqrt{5}}{5}$

Q2. The equation $x^3 - 4x^2 + 7x + 3 = 0$ has roots $x = \alpha$, $x = \beta$ and $x = \gamma$. Which of the equations below has roots $x = -\alpha$, $x = -\beta$ and $x = -\gamma$?

A $x^3 - 4x^2 + 7x + 3 = 0$

B $x^3 + 4x^2 + 7x - 3 = 0$

C $x^3 - 4x^2 + 7x - 3 = 0$

D $x^3 + 4x^2 - 7x - 3 = 0$

Q3. Which of the following is $\int_0^{\frac{\pi}{2}} 2x \cos x \, dx$?

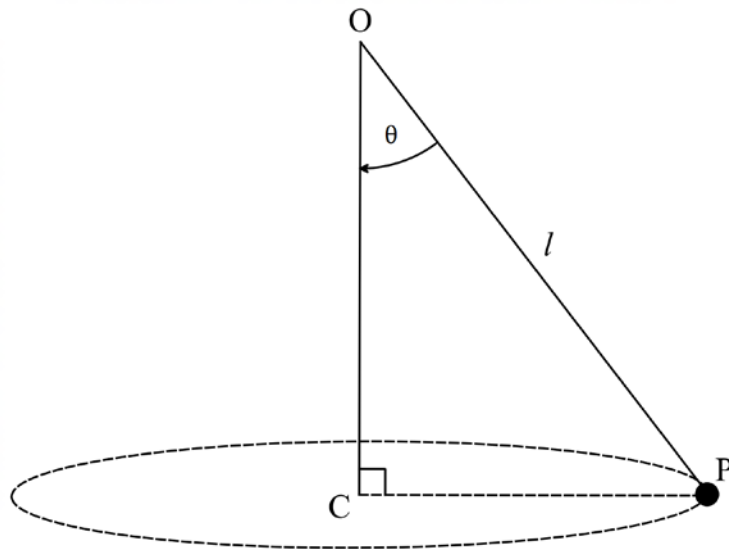
A $\pi - 2$

B $\pi + 2$

C $\frac{\pi^2}{4}$

D $\frac{\pi^2}{8}$

Q4.



The diagram shows a particle P of mass m kilograms suspended from a fixed point O by a light inextensible string of length l metres.

P moves in a circle with centre C directly below O and has uniform angular speed $\omega \text{ rad s}^{-1}$.

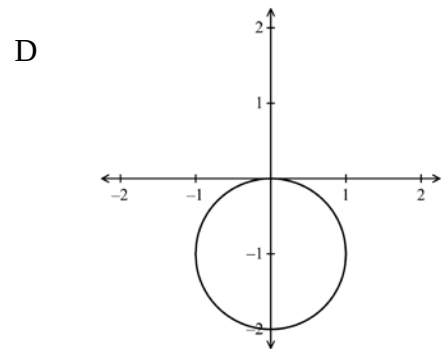
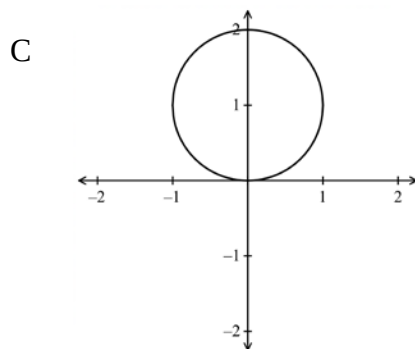
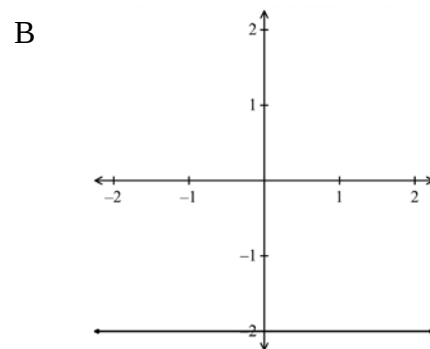
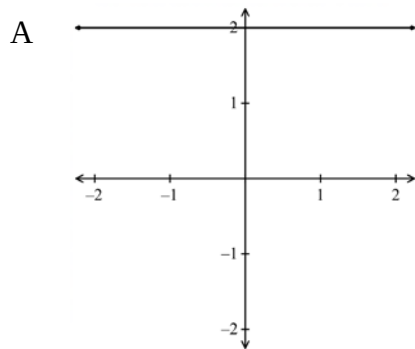
The string makes an angle θ with the vertical line CO and the acceleration due to gravity is $g \text{ ms}^{-2}$.

Which of the following is the tension T in the string?

- A $m l \omega$ newtons
- B $m l \omega^2$ newtons
- C $m g l \omega$ newtons
- D $m g l \omega^2$ newtons

Q5. The complex number z is such that $\operatorname{Im}\left(\frac{1}{z}\right) = -\frac{1}{2}$.

Which of the diagrams below represents the locus of z ?



Q6. The equation $|z - 3| + |z + 3| = 10$ defines an ellipse.

The length of the semi minor axis is:

- A 4
- B 5
- C 8
- D 10

Q7. Consider the following statements:

$$\text{I} \quad \int_0^1 \frac{dx}{1+x^n} < \int_0^1 \frac{dx}{1+x^{n+1}}$$

$$\text{II} \quad \int_0^{\frac{\pi}{2}} \sqrt{\sin x} \, dx = \int_0^{\frac{\pi}{2}} \sqrt{\cos x} \, dx$$

Which of these statements is correct?

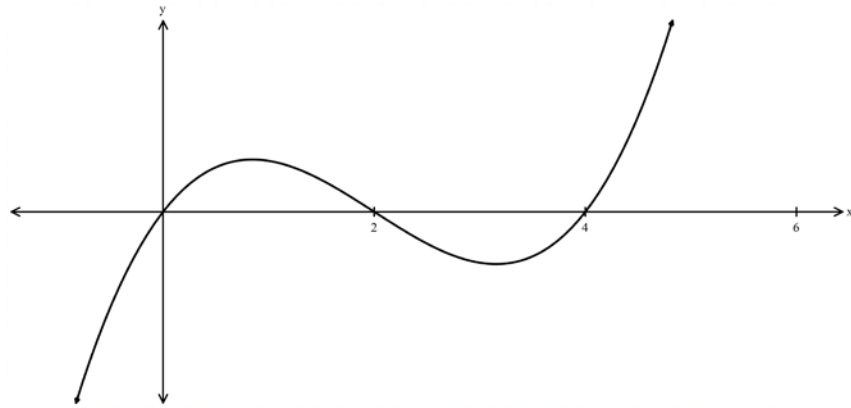
- A Both statements I and II are correct
- B Only statement I is true
- C Only statement II is true
- D Both statements are false

Q8. The hyperbola with equation $xy = 8$ is the hyperbola $x^2 - y^2 = a^2$ referred to different axes.

What is the value of a ?

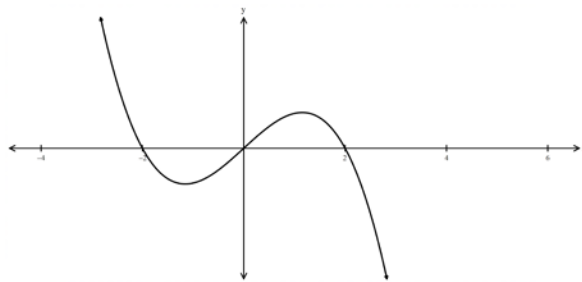
- A 2
- B 4
- C 8
- D 16

Q9.

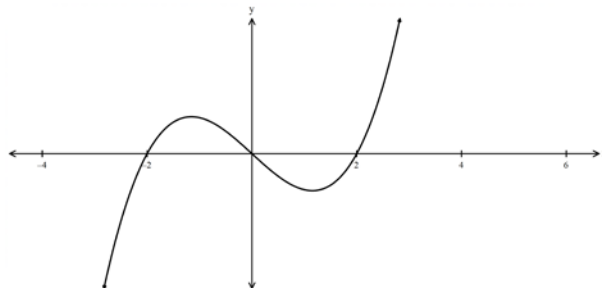


The graph of $y = f(x)$ is shown above. The graph of $y = f(2 - x)$ is:

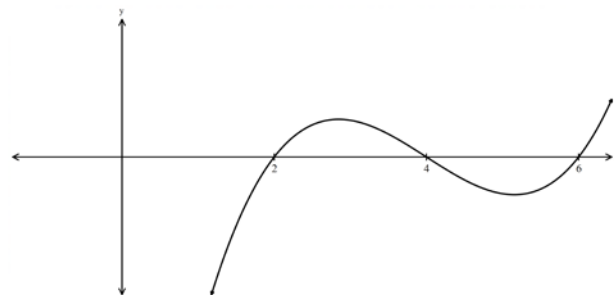
A



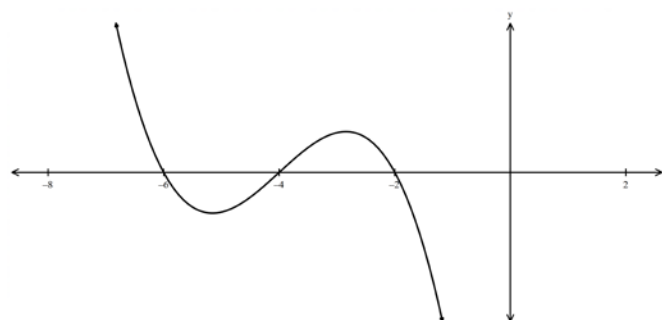
B



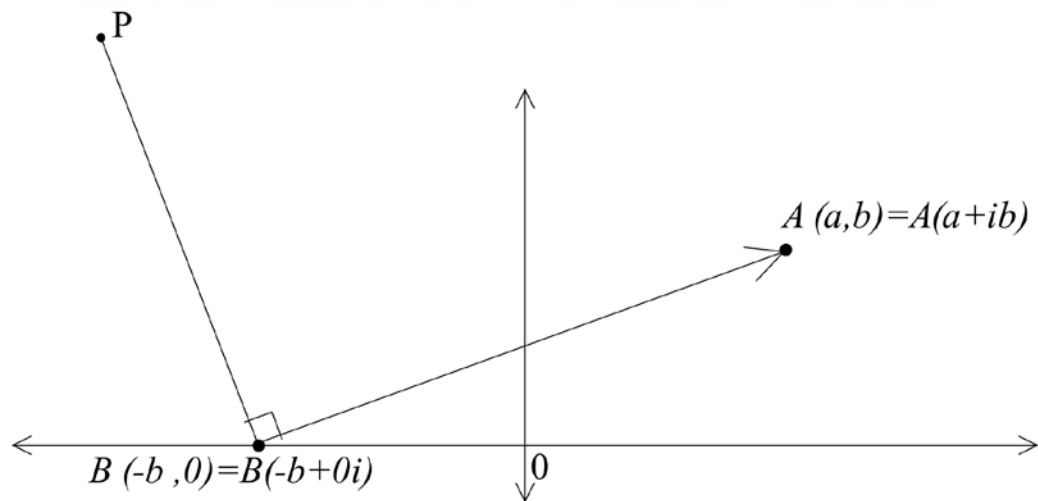
C



D



Q10.



The Argand diagram above shows the point $A(a, b)$ representing the complex number $z = a + ib$, where a and b are real. B is the point $(-b, 0)$.

P is a point such that $PB = 2AB$ and $\angle ABP = 90^\circ$.

Which of the following complex numbers does P represent?

- A $-2b + 2ai$
- B $-b + ai$
- C $-2b + (2a + 2b)i$
- D $-3b + (2a + 2b)i$

End of Multiple Choice

Section II Total Marks is 90

Attempt Questions 11 – 16.

Allow approximately 2 hours & 45 minutes for this section.

Answer all questions, starting each new question in a new booklet with your **student ID number** in the top right hand corner and the question number on the left hand side of your paper.

All necessary working must be shown in each and every question.

Question 11. – Start New Booklet

15 marks

a) Given $z = 1 + i$, represent on an Argand diagram each of the following: 3

i. z

ii. $\frac{1}{z}$

iii. z^2

b) Prove that for any two complex numbers z_1 and z_2 ,

i. $|z_1 - z_2|^2 + |z_1 + z_2|^2 = 2|z_1|^2 + 2|z_2|^2$ 2

ii. Give a geometric interpretation of this result 1

c) Find the equation of the normal to the curve $x^3 - 6xy + y^3 = 5$ at the point $(1, -2)$. 3

d) Show that $\int \frac{\sin^3 x}{\cos^2 x} dx = \sec x + \cos x$ 3

e) The hyperbola H: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (where $a > 0, b > 0$) 3

has a focus at the point $(2\sqrt{13}, 0)$. The line $y = \frac{2}{3}x$ is an asymptote.

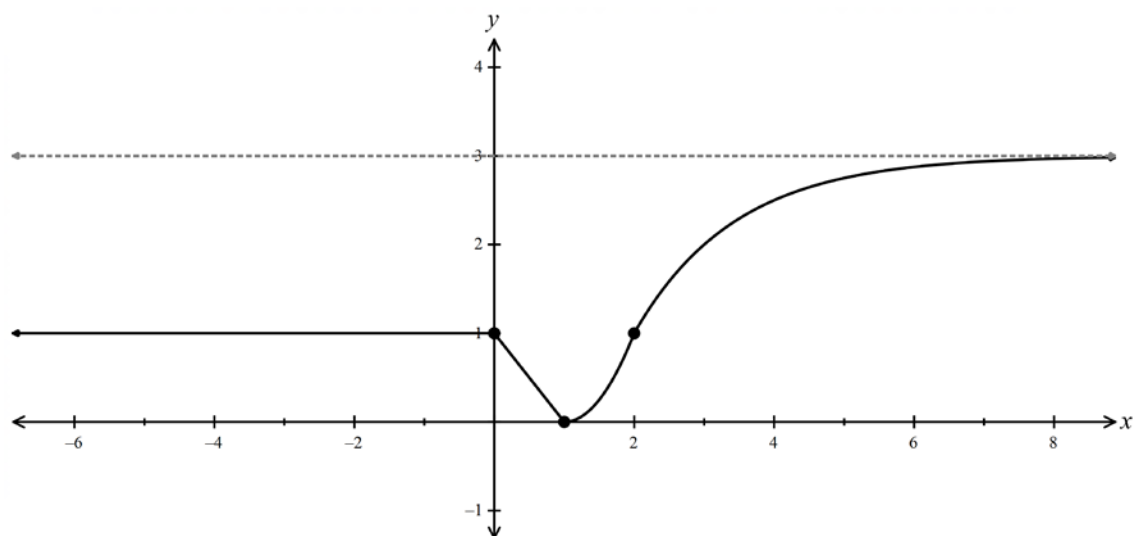
Find the values of a and b .

End of Question 11

Question 12. – Start New Booklet

15 marks

a)



The diagram shows the graph $y = f(x)$.

On separate diagrams, draw $\frac{1}{3}$ page sketches of the following graphs:

(i) $y = f(|2x|)$ 2

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y = \sqrt{f(x)}$ 2

(iv) $y = e^{f(x)}$ 2

(v) $y = x^{f(x)}$ 3

b)

i. Find $\sqrt{-8 + 6i}$ in cartesian form 2

ii. Hence, solve the equation 2

$$2z^2 - (3 + i)z + 2 = 0$$

Express the result in the form $x + iy$

End of Question 12

Question 13. – Start New Booklet

15 marks

a)

i. On the same diagram, sketch the graphs of $y = |x^3 - 1|$ and $y = 1 - x$ 1

ii. Hence, or otherwise, solve $|x^3 - 1| < 1 - x$ 2

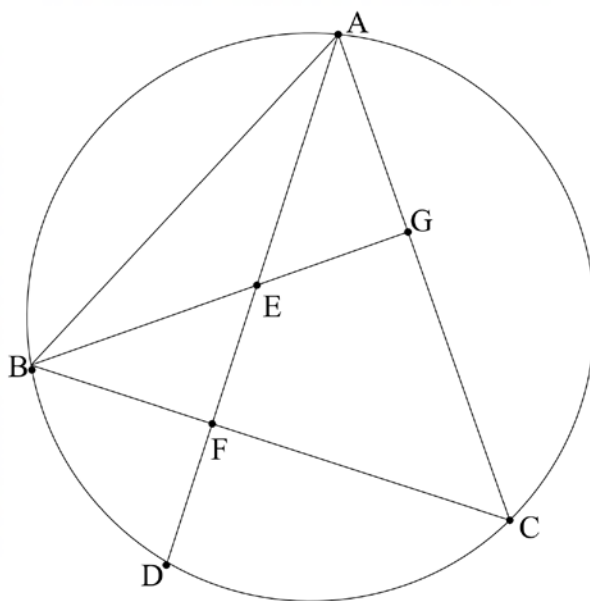
b) Draw neat, labelled sketches to indicate the regions of the Argand plane defined by:

i. $|z| \leq 2$ and $0 \leq \arg z \leq \frac{\pi}{4}$ 2

ii. $|z - \bar{z}| \leq 6$ and $0 \leq \operatorname{Re}(2z) \leq 4$ 2

c) The equation $8x^4 + 44x^3 + 54x^2 + 25x + 4 = 0$ has a triple root. Solve the equation completely. 4

d)



The diagram shows ΔABC inscribed in a circle.

G is the point on AC such that $BG \perp AC$. F is the point on BC such that $AF \perp BC$. AF and BG intersect at E .

AF produced meets the circle at D .

USE THE SUPPLIED SHEET TO ANSWER THE FOLLOWING:

i. Explain why $ABFG$ is a cyclic quadrilateral. 1

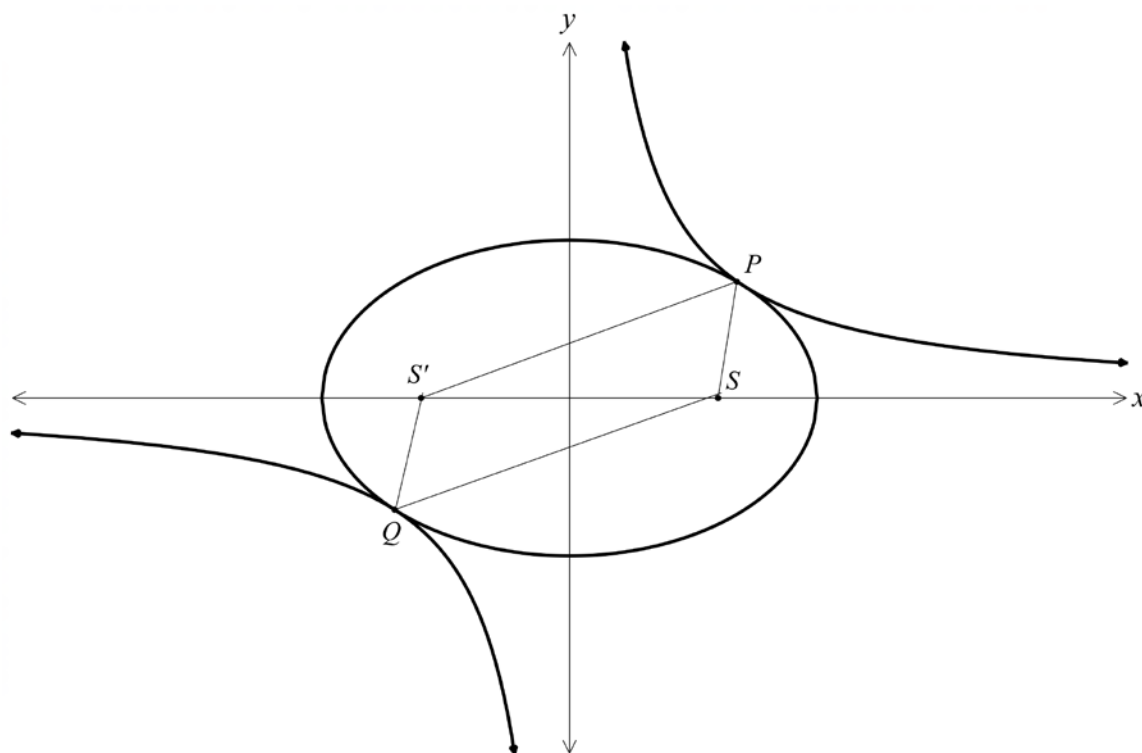
ii. Show that $DF = EF$. 3

End of Question 13

Question 14. – Start New Booklet

15 marks

- a) The ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (where $a > b > 0$) has area $A = \pi ab$, eccentricity e and foci S and S' . The ellipse E touches the hyperbola $H: xy = \frac{1}{2}ab$ at the points P and Q .



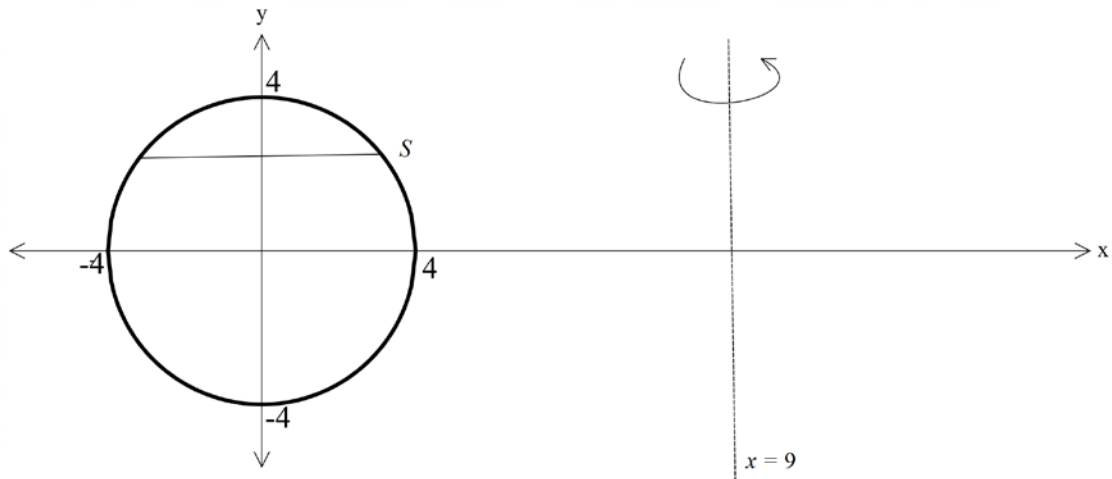
- i. Find the coordinates of P in terms of a and b . 2
 - ii. Show that the ratio of the area of the quadrilateral $PSQS'$ to the area of the ellipse E is $e\sqrt{2} : \pi$ 2
- b) Find $\int \frac{x^2 + 6}{x^2 + x - 6} dx$ 3
- c) Using the result $\int_0^a f(x) dx = \int_0^a f(a-x) dx$, or otherwise, evaluate 3

$$\int_0^{\frac{\pi}{2}} \frac{\cos x + \sin x}{\cos x - \sin x} dx$$

Question 14 continues next page.

Question 14 continued

- d) The circle $x^2 + y^2 = 16$ is rotated about the line $x = 9$ to form a torus.
When the circle is rotated, the line segment S at height y sweeps out an annulus.



- i. Show that the area A of the annulus is given by $A = 36\pi\sqrt{16 - y^2}$ 3
- ii. Find the volume of the torus in exact form. 2

End of Question 14

Question 15. – Start New Booklet**15 marks**

a)

i. Let ω be a complex root of the equation $x^3 = 1$. Show that the other complex root is ω^2 1

ii. Show that $1 + \omega + \omega^2 = 0$ 1

iii. Find the monic cubic equation for which the roots are $\alpha + \beta$, $\alpha\omega + \beta\omega^2$ and $\alpha\omega^2 + \beta\omega$ where α, β are real numbers. 3

b) Using the substitution $t = \tan\left(\frac{x}{4}\right)$, or otherwise, evaluate 4

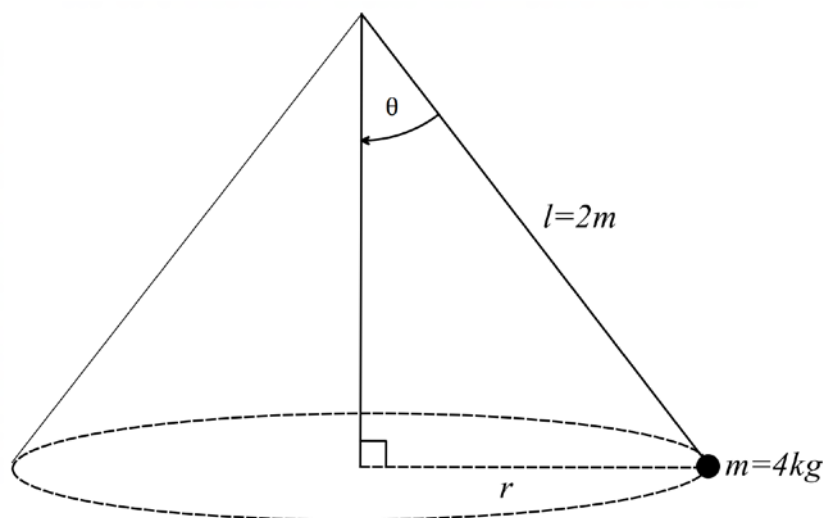
$$\int_0^{\pi} \frac{1}{1 + \cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right)} dx$$

State your answer in simplest exact form.

Question 15 continues next page.

Question 15 continued.

c)



A particle of mass 4kg is attached to a string 2 metres in length.

The particle and string revolve as a conical pendulum.

The constant speed of the particle is $v = \sqrt{g} \text{ ms}^{-1}$, where $g \text{ ms}^{-2}$ is the acceleration due to gravity.

Let θ be the angle of inclination of the string to the vertical, and let r metres be the radius of the horizontal circle in which the particle is revolving, and let T newtons be the tension in the string.

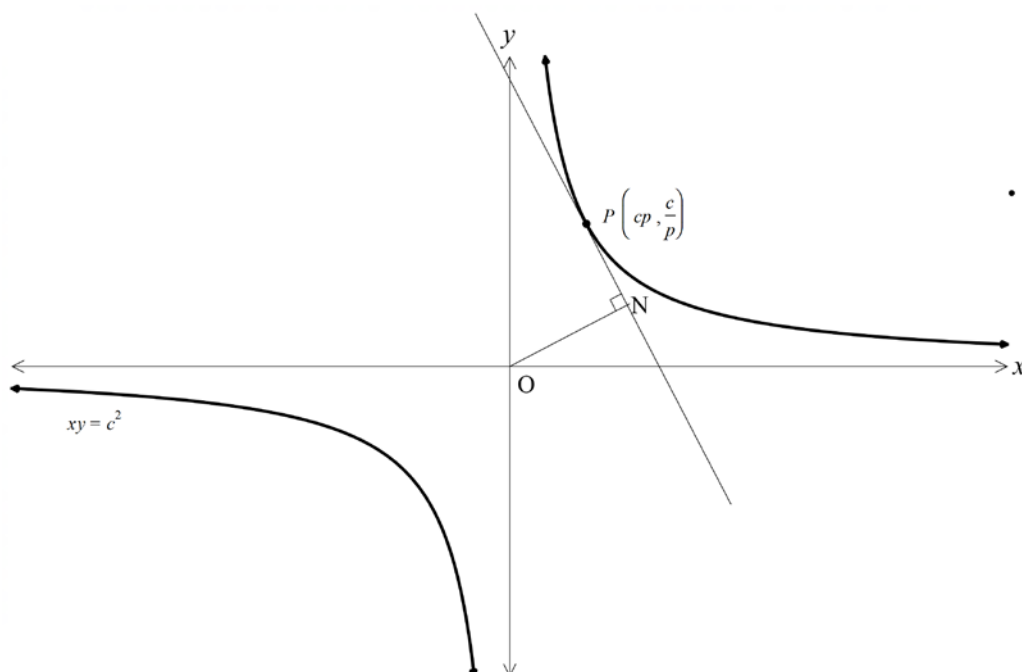
- i. Show that $\tan \theta = \frac{1}{r}$ 2
- ii. Hence show that $\cos \theta = \frac{\sqrt{17} - 1}{4}$ 3
- iii. Find the value of T , correct to one decimal place, given $g = 9.8 \text{ ms}^{-2}$. 1

End of Question 15

Question 16. – Start New Booklet

15 marks

a)



The equation of the tangent to the hyperbola $xy = c^2$ at the point $P\left(cp, \frac{c}{p}\right)$ is

$$x + p^2y = 2cp$$

The point N is the foot of the perpendicular line ON drawn from the origin O to the tangent at P .

(i) Show that the coordinates of N are $\left(\frac{2cp}{1+p^4}, \frac{2cp^3}{1+p^4}\right)$ 2

(ii) Show that the Cartesian equation of the locus of N is $(x^2 + y^2)^2 = 4c^2xy$ 2

Question 16 continues next page.

Question 16 continued.

b)

i. Let $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$ for $n = 0, 1, 2, \dots$ 3

Show that $I_n = \frac{n-1}{n} I_{n-2}$

ii. Show that $\frac{I_{2n}}{I_0} = \frac{(2n)!}{2^{2n} \times (n!)^2}$ for $n = 0, 1, 2, \dots$ 3

iii. Let $y = f(x)$ be a continuous function over $0 \leq x \leq a$. 2

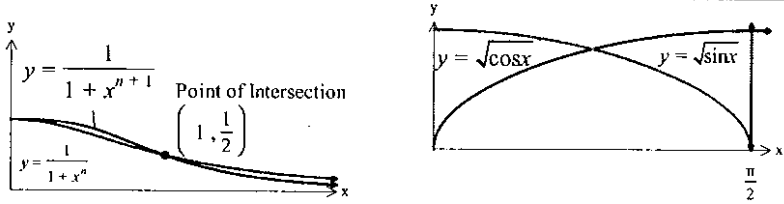
Show that $\int_0^a f(x) \, dx = \int_0^{\frac{a}{2}} \{f(x) + f(a-x)\} \, dx$

iv. Using parts (i) and (iii), show that 3

$$\int_0^{\pi} x \cos^6 x \, dx = \frac{5\pi^2}{32}$$

End of Examination

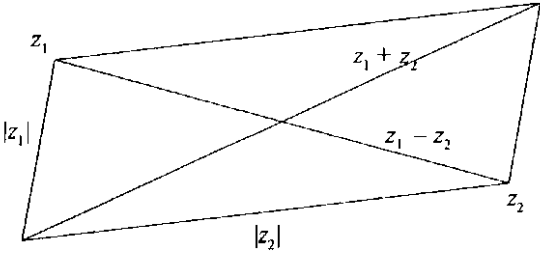
MULTIPLE CHOICE

1	$x^2 - 4y^2 = 4$ $\frac{x^2}{4} - y^2 = 1$ $\text{Distance} = 2 \frac{a}{e}$ $a = 2 \quad b = 1$ $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{\frac{5}{4}}$ $= \frac{\sqrt{5}}{2}$ $D = 2 \times \left(\frac{\sqrt{5}}{2} \right)$ $= \frac{8}{\sqrt{5}} = \frac{8\sqrt{5}}{5}$	D Check.
2		B
3		A
4		B
5		C
6		A
7	 Both are true.	A
8	$xy = c^2 = 8$ $x^2 - y^2 = a^2$ $c^2 = \frac{1}{2} a^2 = 8$ $\therefore a^2 = 16$ $a = 4$	B

9		A
10	$\begin{aligned}\vec{OP} &= \vec{OB} + \vec{BP} \\ &= -b + i(2AB) \\ &= -b + 2i(a + ib - (-b + 0i)) \\ &= -b + 2i(a + b) + 2i^2b \\ &= -3b + 2i(a + b)\end{aligned}$	D

QUESTION 11

a	$z = 1 + i$ $z^2 = 1 + 2i + i^2 = 2i$ $\frac{1}{z} = \frac{1}{1+i} \times \frac{1-i}{1-i}$ $= \frac{1-i}{2}$	3 marks- correct solution for each
bi	$\begin{aligned}& z_1 - z_2 ^2 + z_1 + z_2 ^2 \\ &= (z_1 - z_2)(\overline{z_1 - z_2}) + (z_1 + z_2)(\overline{z_1 + z_2}) \\ &= (z_1 - z_2)\left(\overline{z_1} - \overline{z_2}\right) + (z_1 + z_2)\left(\overline{z_1} + \overline{z_2}\right) \\ &= z_1\overline{z_1} - z_1\overline{z_2} - z_2\overline{z_1} + z_2\overline{z_2} + z_1\overline{z_1} + z_1\overline{z_2} + z_2\overline{z_1} + z_2\overline{z_2} \\ &= z_1 ^2 + z_2 ^2 + z_1 ^2 + z_2 ^2 \\ &= 2 z_1 ^2 + 2 z_2 ^2\end{aligned}$	2 marks- correct solution 1 mark- partial correct expansion and simplification of RHS/LHS

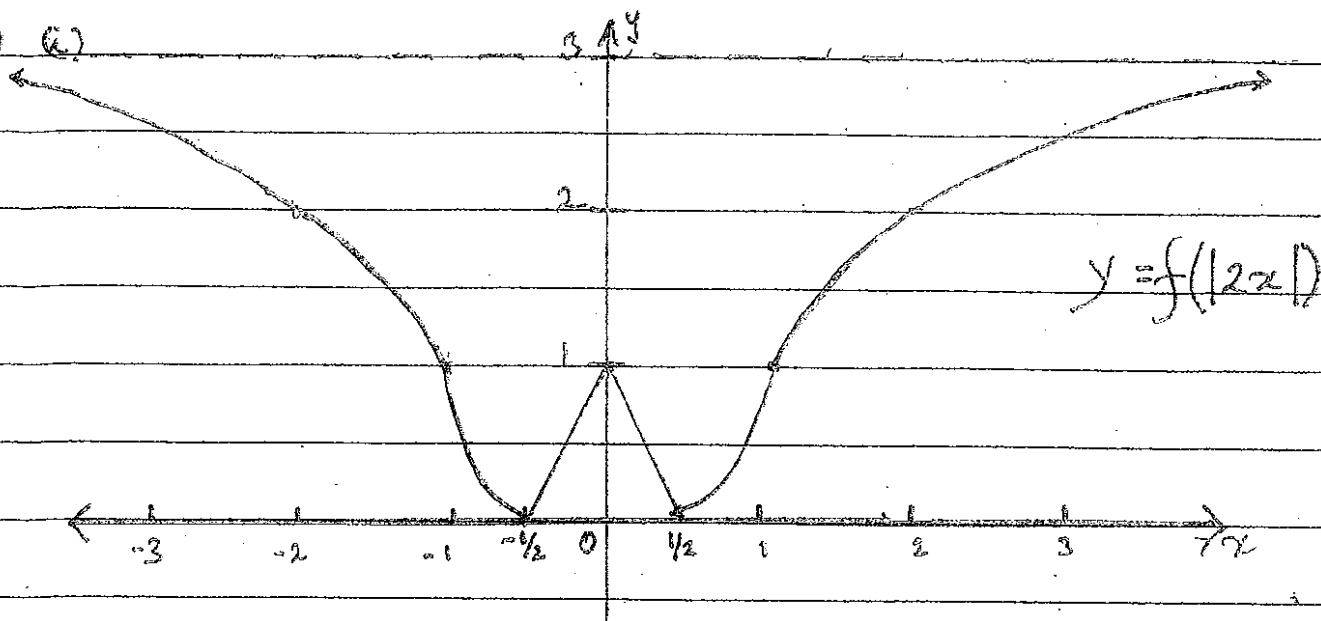
bii	 2 x magnitudes of sides = sum squares of magnitudes of diagonals	1 mark- correct explanation
c	$x^3 - 6xy + y^3 = 5$ $\frac{d}{dx}\{x^3 - 6xy + y^3\} = \frac{d}{dx}\{5\}$ $3x^2 + 3y^2 \frac{dy}{dx} - 6\left\{\frac{xdy}{dx} + y\right\} = 0$ $x^2 + y^2 \frac{dy}{dx} - 2x \frac{dy}{dx} - 2y = 0$ $\frac{dy}{dx}\{y^2 - 2x\} = 2y - x^2$ $\frac{dy}{dx} = \frac{2y - x^2}{y^2} - 2x$ at (1, -2) $m_1 = \frac{-4 - 1}{(-2)^2 - 2(1)}$ $= -\frac{5}{2}$ $m_2 = \frac{2}{5}$ $y + 2 + \frac{2}{5}(x - 1)$ $2x - 5y - 12 = 0$	3 marks- correct solution 2 marks- partial correct with only one error 1 mark- correct application to eqn of line but incorrect diff and gradient

d	$\int \frac{\sin^3 x}{\cos^2 x}$ $= \int \sin x \left\{ \frac{1 - \cos^2 x}{\cos^2 x} \right\} dx$ $= \int \sin x \cos^2 x \, dx - \int \frac{\sin x \cos^2 x}{\cos^2 x} dx$ $= \sin x (\cos x)^{-2} dx - \int \sin x \, dx$ $= - \int (-\sin x) (\cos x)^{-2} dx - \int \sin x \, dx$ $= - \frac{(\cos x)^{-1}}{-1} - (-\cos x)$ $= \frac{1}{\cos x} + \cos x$ $= \sec x + \cos x$	3 marks- correct solution 2 marks- partial correct with only one error 1 mark- one correct technique in relevant progress to int
e	$H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ $S(ae, 0) \Rightarrow S(2\sqrt{13}, 0) \Rightarrow ae = 2\sqrt{13}$ $y = \frac{b}{a}x \Rightarrow y = \frac{2}{3}x \Rightarrow \frac{b}{a} = \frac{2}{3} \Rightarrow \frac{b^2}{a^2} = \frac{4}{9}$ $a^2 e^2 = 4 \times 13$ $a^2 \left[1 + \frac{b^2}{a^2} \right] = 52$ $a^2 \left[1 + \frac{4}{9} \right] = 52$ $a^2 = 36$ $a = 6$ $\frac{b}{6} = \frac{2}{3}$ $b = 4$	3 marks- correct solution 2 marks- partial correct only one error in correct progress 1 mark- correct solution for values of a, b and e

Question 12

1

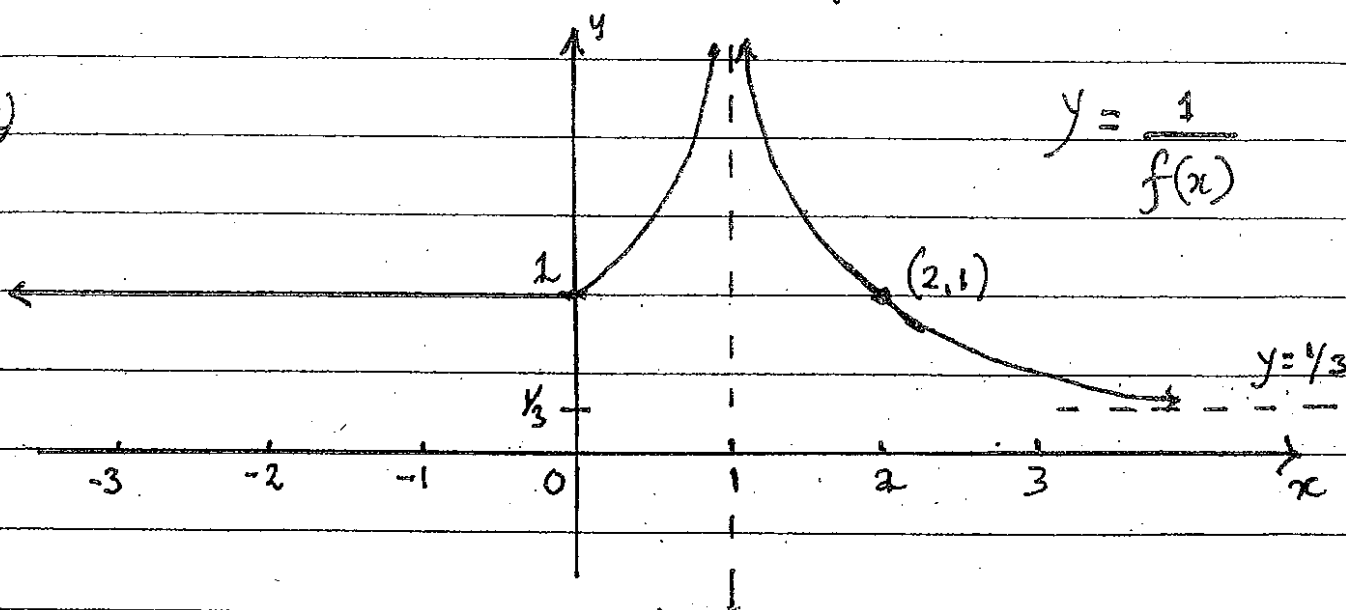
a) (i)



2 Marks: correct answer

1 Mark: correct shape

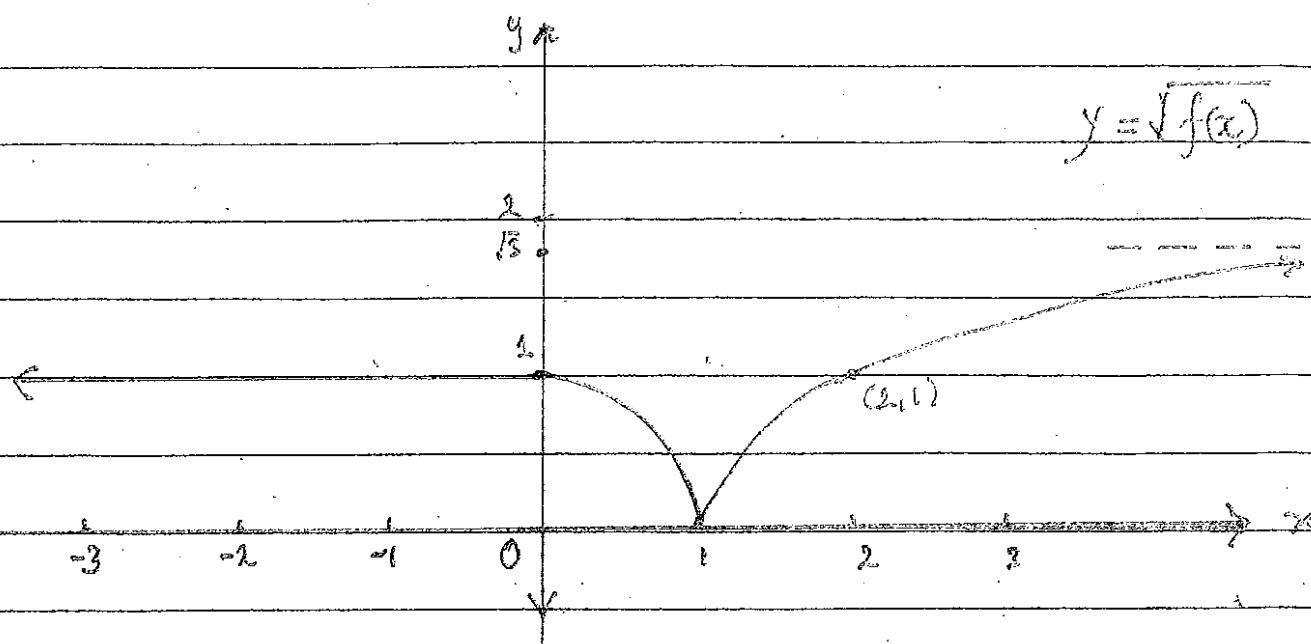
(ii)



2 Marks: correct answer

1 Mark: correct shape

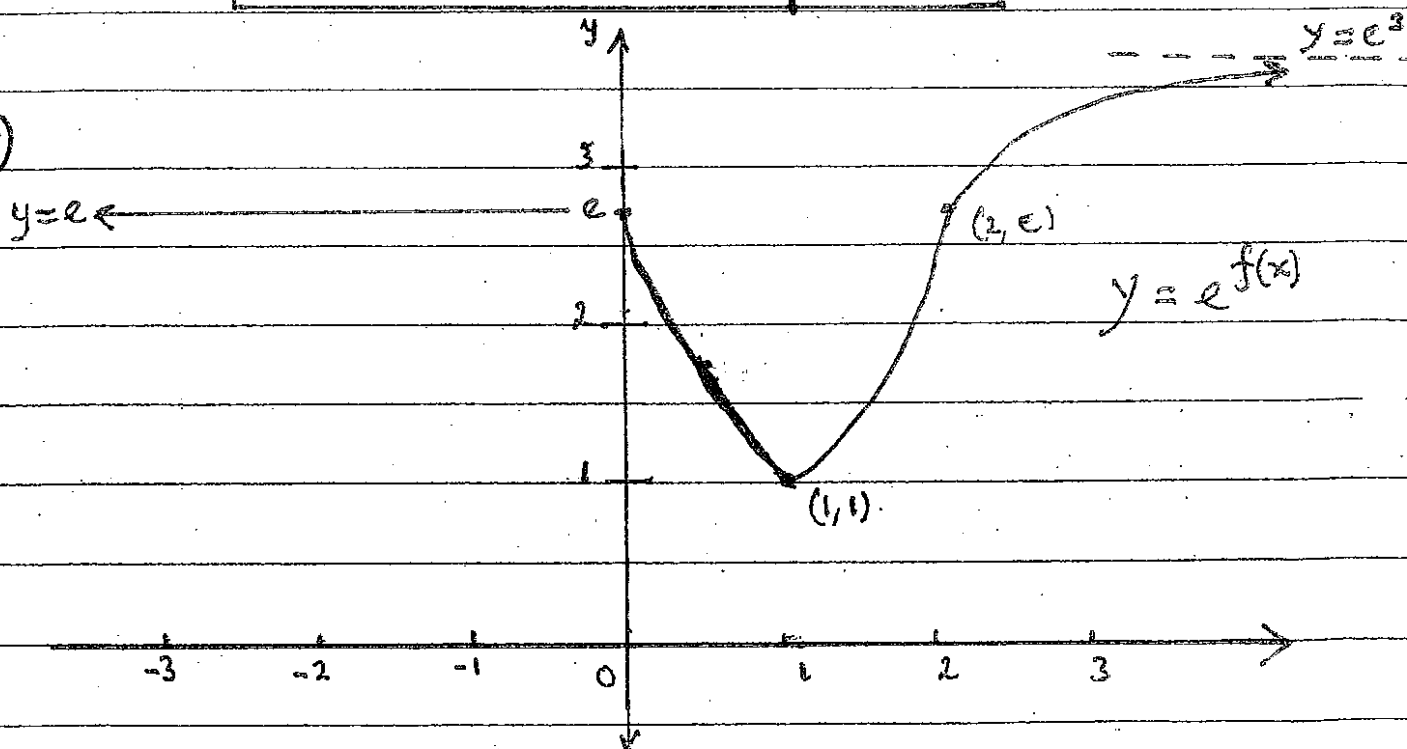
(iii)



2 Marks: correct answer

1 Mark: correct shape

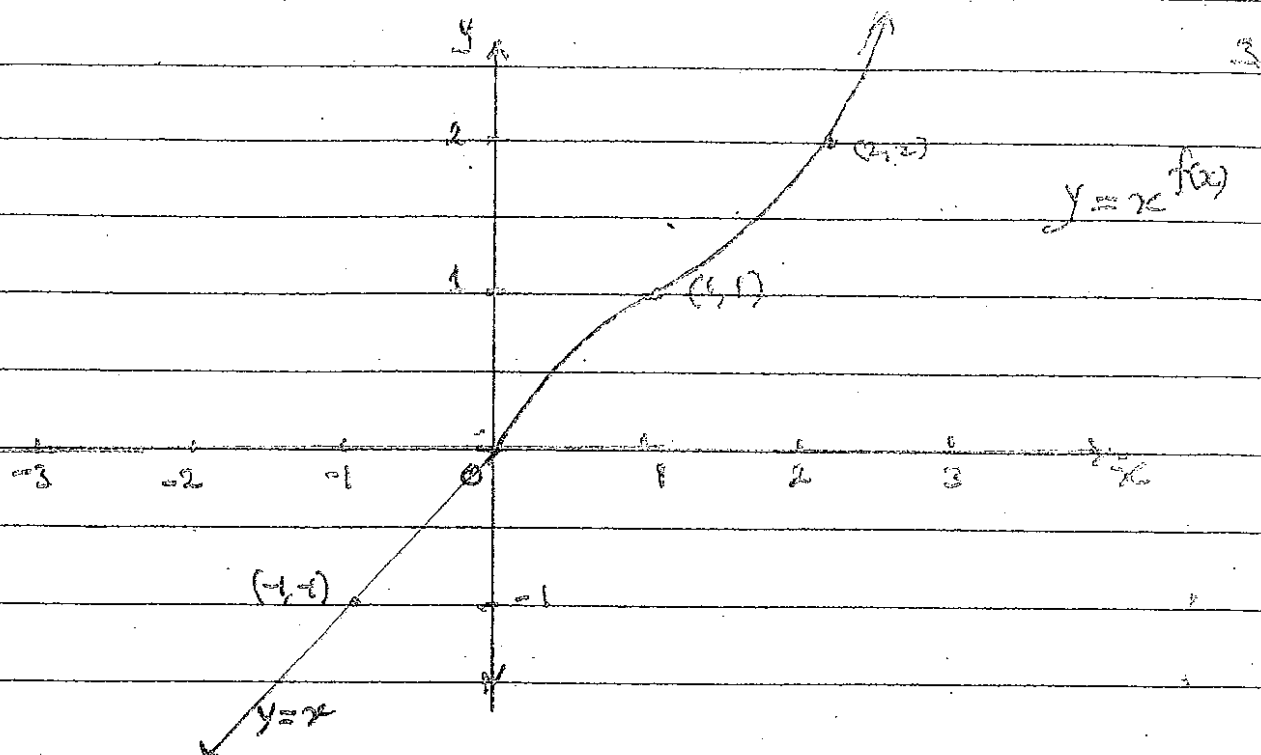
(iv)



2 Marks: correct answer

1 Mark: correct shape

(v)



3 Marks: correct answer

2 Marks: two correct parts

1 Mark: one correct part

$$(b) (i) \quad z = x + iy = \sqrt{-8 + 6i}$$

$$x^2 - y^2 + i(2xy) = -8 + 6i$$

$$x^2 - y^2 = -8 \quad xy = 3 \Rightarrow y = 3/x$$

$$x^2 - (3/x)^2 = -8$$

$$\Rightarrow x^4 + 8x^2 - 9 = 0$$

$$(x^2 + 9)(x^2 - 1) = 0$$

$$x = \pm 1 \Rightarrow y = \pm 3$$

$$z = \pm 1 \pm 3i$$

2 Marks: correct solution
(other methods possible)1 Mark: significant progress
depending on method

$$\sqrt{-8 + 6i} = 1 + 3i$$

(ii)

$$2z^2 - (3+i)z + 2 = 0$$

$$z = \frac{(3+i) \pm \sqrt{(3+i)^2 - 4 \times 2 \times 2}}{2 \times 2}$$

$$z = \frac{(3+i) \pm \sqrt{-8+6i}}{4}$$

$$z_1 = \frac{3+i - (1+3i)}{4}$$

$$= \frac{2-2i}{4}$$

$$z_1 = \frac{1-i}{2}$$

$$z_2 = \frac{3+i + 1+3i}{4}$$

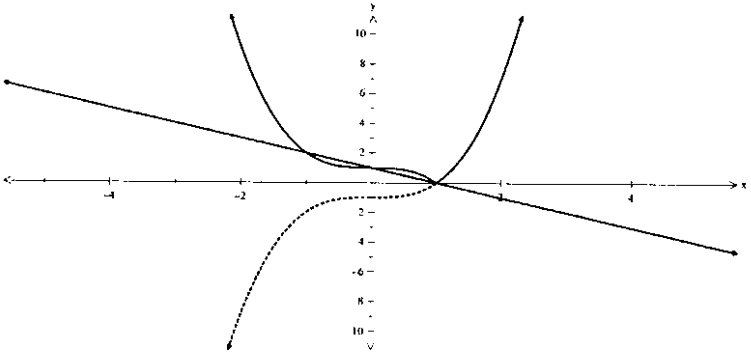
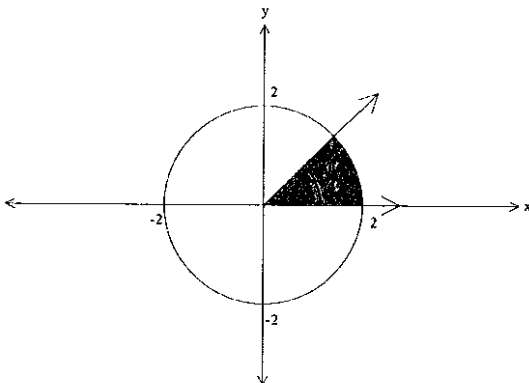
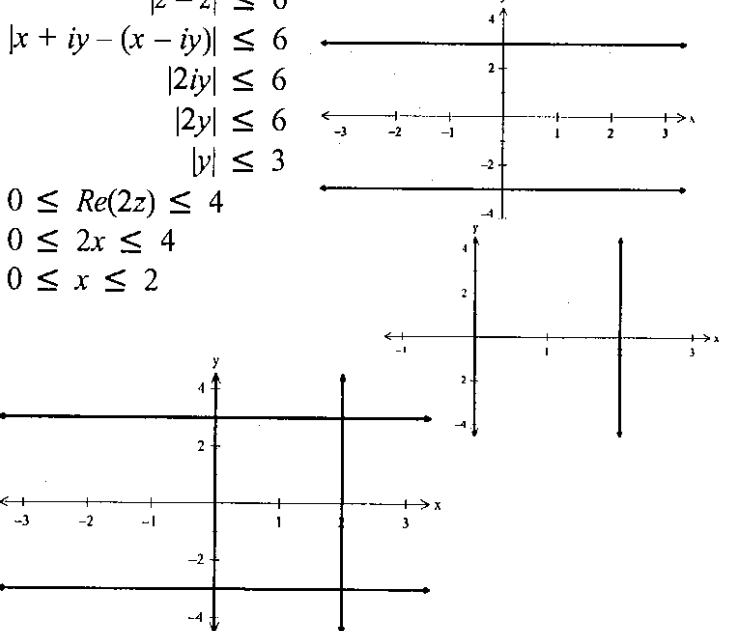
$$= \frac{4+4i}{4}$$

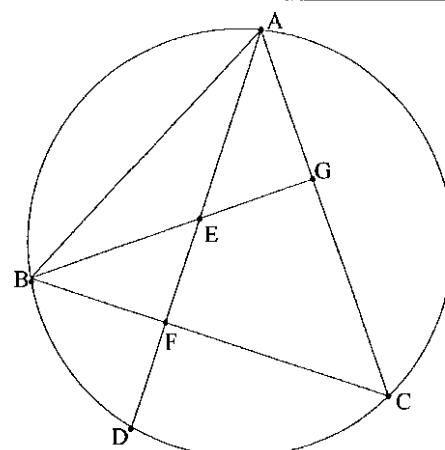
$$z_2 = 1+i$$

2 Marks: correct answer

1 Mark: one correct value, z_1
or z_2

QUESTION 13

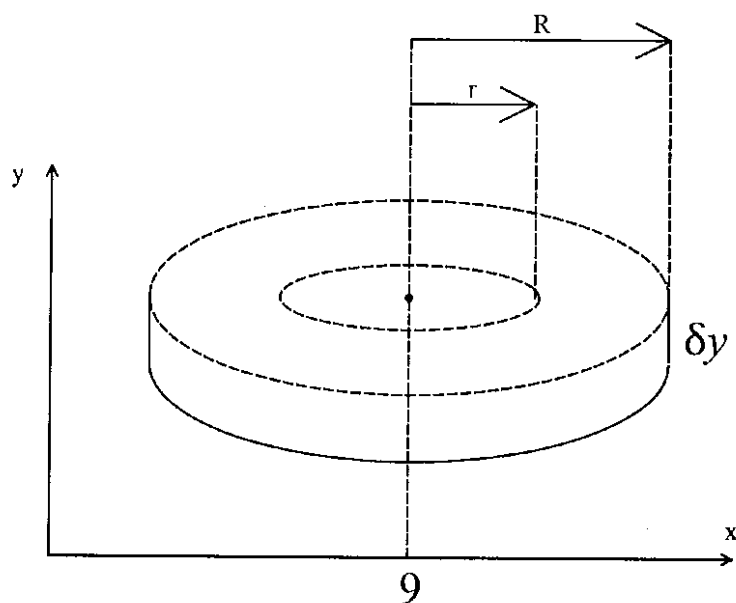
ai		1 mark- correct solution
aii	$-(x^3 - 1) = 1 - x$ $1 - x^3 = 1 - x$ $(1 - x)(1 + x + x^2) = 1 - x$ $1 + x + x^2 = 1, x \neq 1$ $x(x + 1) = 0$ $x = 0, -1$ $\therefore x^3 - 1 < 1 - x$ $-1 < x < 0$	2 marks- correct solution 1 mark- correct solution from incorrect graph or only partial correct region (only one error)
bi		2 marks- correct solution 1 mark- partial correct region (only one error)
bii	$ z - \bar{z} \leq 6$ $ x + iy - (x - iy) \leq 6$ $ 2iy \leq 6$ $ 2y \leq 6$ $ y \leq 3$ $0 \leq \operatorname{Re}(2z) \leq 4$ $0 \leq 2x \leq 4$ $0 \leq x \leq 2$ 	2 marks- correct solution 1 mark- partial correct region (only one error)

c	$P(x) = 8x^4 + 44x^3 + 54x^2 + 25x + 4$ $P'(x) = 32x^3 + 132x^2 + 108x + 25$ $P''(x) = 96x^2 + 264x + 108$ $\therefore 24x^2 + 66x + 27 = 0$ $x = \frac{-66(\pm\sqrt{66^2 - (4 \times 24 \times 27)})}{48}$ $= -\frac{9}{4}, -\frac{1}{2}$ $P'\left(-\frac{9}{4}\right) = \frac{343}{4}$ $P'\left(-\frac{1}{2}\right) = 0$ $\therefore x = -\frac{1}{2} \Rightarrow 2x + 1 = 0$ $P(x) = (2x + 1)^3(\alpha x + \beta)$ $\alpha = 1, \beta = 4$ $P(x) = (2x + 1)^3(x + 4)$ $x = -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -4$ $P(-4) = 0$	4 marks- correct solution 3 marks- correct solution from incorrect triple root 2 marks- partial correct with solution to triple root and check 1 mark- only correctly solves for values of x on $P''(x)$
di	 $\angle AGB = 90^\circ$ (given) $\angle AFB = 90^\circ$ (given) $\therefore \angle AFB = \angle AGB$ $\therefore ABFG$ is cyclic (= $\angle$ on chord AB)	1 mark- correct solution with correct reasons
dii	join AB $\angle DAC = \angle DBC = x$ (= $\angle$ on "arc CD circle ABDC") ang FAG = ang FBG = x (= ang on "arc FG circle ABFG") $\therefore \angle DBF = \angle EBF = x$ $\triangle DBE$ is isos ($\perp$ to Δ bisects $\angle DBE$) $DE = DF$ ($\perp$ from apex to base bisects "DE") OR $\triangle DBF \equiv \triangle EBF$ (AAS) $DE = DF$ (corr sides in congruent Δ)	3 marks- correct solution with correct reasons 2 marks- only one error at least two correct and relevant theorems 1 mark- one correct and relevant theorem

QUESTION 14

a-i	$xy = \frac{1}{2}ab$ $y = \frac{ab}{2x} \Rightarrow y^2 = \frac{a^2b^2}{4x^2}$ $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\frac{x^2}{a^2} + \frac{1}{b^2} \times \frac{a^2b^2}{4x^2} = 1$ $\frac{x^2}{a^2} + \frac{a^2}{4x^2} = 1$ $4x^4 + a^4 = 4x^2a^2$ $4x^4 - 4x^2a^2 + a^4 = 0$ $(2x^2 - a^2)^2 = 0$ $x^2 = \frac{a^2}{2}$ $x = \pm \frac{a}{\sqrt{2}}$ $x = \frac{a}{\sqrt{2}} \Rightarrow y = \frac{ab}{2} \times \frac{\sqrt{2}}{a}$ $P = \left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}} \right)$ $Q = \left(-\frac{a}{\sqrt{2}}, -\frac{b}{\sqrt{2}} \right)$	2 marks
a-ii	distance $S'S' = 2ae$ Area of $\Delta S'SP = \frac{1}{2} \times 2ae \times \frac{b}{\sqrt{2}}$ $= \frac{a b e}{\sqrt{2}}$ $\therefore$ quad $QS'PS$ $= 2 \times \frac{a b e}{\sqrt{2}}$ $= \sqrt{2} a b e$ $\frac{\text{Area } QS'PS}{\text{Area } E} = \frac{\sqrt{2} a b e}{\pi ab} = \frac{\sqrt{2} e}{\pi}$ $\therefore$ Ratio $\Rightarrow \sqrt{2} e : \pi$	2 marks – correct solution for general case 1 mark - Correct solution for specific solution only - finding a ratio in e only

b	$\int \frac{x^2 + 6}{x^2 + x - 6} dx$ $\frac{x^2 + 6}{x^2 + x - 6} = \frac{x^2 + x - 6 - x + 12}{x^2 + x - 6}$ $= 1 - \frac{x - 12}{x^2 + x - 6}$ $\therefore \frac{x - 12}{x^2 + x - 6} = \frac{A}{x + 3} + \frac{B}{x - 2}$ $x - 12 = A(x - 2) + B(x + 3)$ Let $x = 2$ $-10 = 5B \Rightarrow B = -2$ Let $x = -3$ $-15 = -5A \Rightarrow A = 3$ $\int \frac{x^2 + 6}{x^2 + x - 6} dx$ $= \int 1 - \left[\frac{3}{x + 3} - \frac{2}{x - 2} \right] dx$ $= x - 3 \ln x + 3 + 2 \ln x - 2 $ $= x + \ln \left \frac{(x - 2)^2}{(x + 3)^3} \right + C$	3 marks – correct solution 2 marks – correct use of partial fractions 1 mark – correct rearrangement of initial algebraic fraction
c	$\int_0^{\frac{\pi}{2}} \frac{\cos x + \sin x}{\cos x - \sin x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos\left(\frac{\pi}{2} - x\right) + \sin\left(\frac{\pi}{2} - x\right)}{\cos\left(\frac{\pi}{2} - x\right) - \sin\left(\frac{\pi}{2} - x\right)} dx$ $= \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sin x - \cos x} dx$ $\therefore 2 \times \int_0^{\frac{\pi}{2}} \frac{\cos x + \sin x}{\cos x - \sin x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos x + \sin x}{\cos x - \sin x} dx + \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sin x - \cos x} dx$ $= \int_0^{\frac{\pi}{2}} \frac{\cos x + \sin x}{\cos x - \sin x} dx - \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\cos x - \sin x} dx$ $2I = 0$ $\therefore I = 0$	3 marks – correct solution fully demonstrated. 2 marks – use of result but failure to achieve the correct solution and/or insufficient demonstration 1 mark – use of correct trig identity to simplify. Nb question did require the use of the specified result



2 marks

$$x^2 + y^2 = 16$$

$$x = \pm\sqrt{16 - y^2}$$

$$\therefore R = 9 + \sqrt{16 - y^2}$$

$$r = 9 - \sqrt{16 - y^2}$$

$$\begin{aligned} A &= \pi(R^2 - r^2) \\ &= \pi(R + r)(R - r) \\ &= \pi(18)(2\sqrt{16 - y^2}) \\ &= 36\pi(\sqrt{16 - y^2}) \end{aligned}$$

$$\delta V = A \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_{-4}^4 A \delta y$$

$$= 36\pi \int_{-4}^4 \sqrt{16 - y^2} dy$$

$$= 36\pi \times \text{semicircle}$$

$$= 36\pi \times \frac{1}{2} \pi 4^2$$

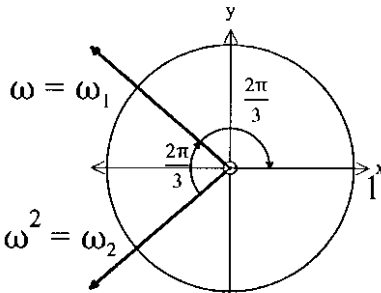
$$= 288\pi^2 \text{ units}^3$$

3 marks – correct solution

2 marks –

1 mark

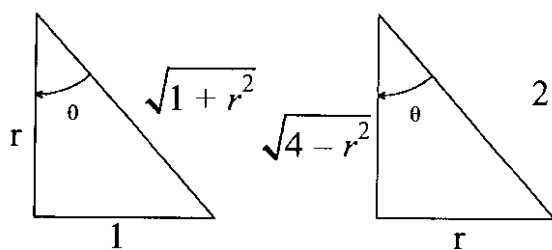
QUESTION 15

a-i	$\omega = \text{cis}\left(\frac{2\pi}{3}\right)$ $\omega_2 = \text{cis}\left(\frac{4\pi}{3}\right) = \left(\text{cis}\left(\frac{2\pi}{3}\right)\right)^2 = \omega^2$ or $\omega^3 = 1$ $\therefore (\omega^3)^2 = 1^2 = 1$ $\therefore (\omega^2)^3 = 1$ $\therefore \omega^2 \text{ is a root}$ 	1 mark – correct explanation
a-ii	Roots of the Polynomial are $1, \omega, \omega^2$ $\Sigma \alpha = -\frac{b}{a} = 0$ $\therefore 1 + \omega + \omega^2 = 0$	1 mark – correct explanation
a-iii	$P(x) = x^3 + bx^2 + cx + d$ $\Sigma \alpha = -\frac{b}{a} = -b \text{ as } a = 1$ $= \alpha + \beta + a\omega + b\omega^2 + a\omega^2 + b\omega$ $= \alpha(1 + \omega + \omega^2) + \beta(1 + \omega + \omega^2)$ $= \alpha \times 0 + \beta \times 0 = 0$ $\therefore b = 0$ $\Sigma \alpha \beta = \frac{c}{a} = c \text{ as } a = 1$ $= (\alpha + \beta)(\alpha\omega + \beta\omega^2) + (\alpha + \beta)(a\omega^2 + \beta\omega) + (\alpha\omega + \beta\omega^2)(\alpha\omega^2 + \beta\omega)$ $= (\alpha^2\omega + \alpha\beta\omega + \alpha\beta\omega^2 + \beta^2\omega^2) + (\alpha^2\omega^2 + \alpha\beta\omega + \alpha\beta\omega^2 + \beta^2\omega)$ $+ (\alpha^2\omega^3 + \alpha\beta\omega^2 + \alpha\beta\omega^4 + \beta^2\omega^3)$ $= \alpha^2(\omega + \omega^2 + \omega^3) + \beta^2(\omega + \omega^2 + \omega^3) + 3\alpha\beta(\omega + \omega^2)^*$ *($\alpha\beta\omega^4 = \alpha\beta\omega^3\omega = \alpha\beta\omega$) also $\omega + \omega^2 = -1$ $c = 0 + 0 - 3\alpha\beta = -3\alpha\beta$	1 mark per correct pronumeral.

	$\Sigma \alpha \beta \gamma = -\frac{d}{a} = -d$ $ \begin{aligned} -d &= (\alpha + \beta)(\alpha\omega + \beta\omega^2)(\alpha\omega^2 + \beta\omega) \\ &= (\alpha + \beta)(\alpha^2\omega^3 + \alpha\beta\omega^4 + \alpha\beta\omega^2 + \beta^2\omega^3) \\ &= (\alpha + \beta)(\alpha^2 + \beta^2 + \alpha\beta(\omega^4 + \omega^2)) \\ &= (\alpha + \beta)(\alpha^2 + \beta^2 + \alpha\beta(\omega + \omega^2)) \\ &= (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta) \\ &= (\alpha^3 + \beta^3) \end{aligned} $ $ \begin{aligned} \therefore P(x) &= x^3 + 0x^2 - 3\alpha\beta x - (\alpha^3 + \beta^3) \\ &= x^3 - 3\alpha\beta x - (\alpha^3 + \beta^3) \end{aligned} $	
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	$\tan\left(\frac{x}{4}\right) = t \quad x = 0 \Rightarrow t = \tan 0 = 0 \quad t = \tan\left(\frac{x}{4}\right)$ $\cos\left(\frac{x}{2}\right) = \frac{1-t^2}{1+t^2} \quad x = \pi \Rightarrow t = \tan\frac{\pi}{4} = 1 \quad x = 4\tan^{-1}(t)$ $\sin\left(\frac{x}{2}\right) = \frac{2t}{1+t^2} \quad dx = \frac{4}{1+t^2} dt$	4 marks – correct solution
b	$\int_0^1 \frac{1}{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \cdot \frac{4}{1+t^2} dt$ $= \int_0^1 \frac{1+t^2}{1+t^2+1-t^2+2t} \times \frac{4}{1+t^2} dt$ $= \int_0^1 \frac{4}{2+2t} dt$ $= 2 \int_0^1 \frac{1}{1+t} dt$ $= 2 [\ln(1+t)]_0^1$ $= 2 \{\ln 2 - \ln 1\}$ $= 2 \ln 2 = \ln 4$	3 marks - incorrect final simplification - one error in final integration process 2 - correct t substitution and limits 1 - correct t substitution or new limits

c-i	$T_y = T \cos \theta = 4g$ $T_x = T \sin \theta = \frac{mv^2}{r} = \frac{4(\sqrt{g})^2}{r} = \frac{4g}{r}$ $\frac{T \sin \theta}{T \cos \theta} = \frac{4g}{r} \div 4g = \frac{1}{r}$	2 – fully demonstrated 1 mark – limited demonstration
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$$\cos\theta = \frac{r}{\sqrt{1+r^2}} \text{ also } \cos\theta = \frac{\sqrt{4-r^2}}{2}$$

$$\therefore \frac{r}{\sqrt{1+r^2}} = \frac{\sqrt{4-r^2}}{2}$$

$$\frac{r^2}{1+r^2} = \frac{4-r^2}{4}$$

$$4r^2 = (1+r^2)(4-r^2)$$

$$4r^2 = 4 + 4r^2 - r^2 - 4r^4$$

$$4r^4 + r^2 - 4 = 0$$

$$r^2 = \frac{-1 \pm \sqrt{17}}{2}$$

$$\text{but } r^2 \geq 0$$

$$r = \frac{-1 + \sqrt{17}}{2}$$

$$\cos\theta = \frac{r}{\sqrt{1+r^2}}$$

$$\cos^2\theta = \frac{r^2}{1+r^2}$$

$$\frac{\sqrt{17}-1}{2}$$

$$= \frac{1 + \frac{\sqrt{17}-1}{2}}{1 + \frac{\sqrt{17}-1}{2}}$$

$$= \frac{\sqrt{17}-1}{\sqrt{17}+1}$$

$$= \frac{(\sqrt{17}-1)^2}{17-1}$$

$$\therefore \cos\theta = \sqrt{\frac{(\sqrt{17}-1)^2}{16}}$$

$$= \frac{\sqrt{17}-1}{4}$$

4marks – correct solution fully demonstrated.

3 marks

- derivation of positive r with reasons

2 marks – 2 correct expressions for $\cos\theta$ and production of quadratic or similar

1 marks – 2 correct expressions for $\cos\theta$

c-ii

c-iii	$T \cos \theta = 4g$ $T \left[\frac{\sqrt{17} - 1}{4} \right] = 4g$ $T = \frac{16 \times 9.8}{\sqrt{17} - 1}$ $= 50.206 \dots$ $\cong 50.2 \text{ N}$	1 mark – correct answer
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Question 16

1

(a) (i) tangent at P: $x + p^2 y = 2cp$

$$\Rightarrow y = -\frac{1}{p^2}x + \frac{2c}{p}$$

$$m_1 = -1/p^2$$

 $\Rightarrow$ gradient of ON, $m_2 = p^2$ equation of ON: $y = p^2 x$

$$\Rightarrow x + p^2(p^2 x) = 2cp$$

$$x(1 + p^4) = 2cp$$

$$x = \frac{2cp}{1 + p^4}$$

$$\underline{\underline{\underline{\frac{2cp}{1 + p^4}}}}}$$

$$\Rightarrow y = p^2 \left(\frac{2cp}{1 + p^4} \right)$$

$$y = \frac{2cp^3}{1 + p^4}$$

$$\underline{\underline{\underline{\frac{2cp^3}{1 + p^4}}}}}$$

$$\therefore N \text{ is } \left\{ \frac{2cp}{1 + p^4}, \frac{2cp^3}{1 + p^4} \right\}$$

2 Marks: correct answer

1 Mark: correct derivation of
either x or y
coordinate.

$$(1) \quad x = \frac{2cp}{1+p^4} \Rightarrow x(1+p^4) = 2cp \Rightarrow x^2(1+p^4)^2 = 4c^2p^2 \quad (1)$$

$$y = p^2x \Rightarrow p^2 = y/x \Rightarrow p^4 = y^2/x^2 \quad (2)$$

Substitute (2) into (1):

$$x^2 \left(1 + \left(\frac{y}{x} \right)^2 \right)^2 = 4c^2 \left(\frac{y}{x} \right)$$

$$x^2 \left(\frac{x^2 + y^2}{x^2} \right)^2 = \frac{4c^2y}{x}$$

$$x^2 \left(\frac{x^2 + y^2}{x^2} \right)^2 = \frac{4c^2y}{x}$$

$$\frac{[x^2 + y^2]^2}{x^2} = \frac{4c^2y}{x}$$

$\times x^2$

$$\underline{\underline{[x^2 + y^2]^2 = 4c^2xy}}$$

2 Marks: correct answer

1 Mark: significant progress
beyond elimination
of p .

$$b) i) I_n = \int_0^{\pi/2} \cos^n x \, dx = \int_0^{\pi/2} \cos^{n-1} x \cdot \cos x \, dx$$

$$u = \cos^{n-1} x$$

$$dv = \cos x$$

$$du = (n-1) \cos^{n-2} x \cdot -\sin x$$

$$v = \sin x$$

$$= -(n-1) \sin x \cos^{n-2} x$$

①

$$I_n = \left[\sin x \cos^{n-1} x \right]_0^{\pi/2} - \int_0^{\pi/2} -(n-1) \cos^{n-2} x \sin x \cdot \sin x \, dx$$

$$= \left[0 - 0 \right] + (n-1) \int_0^{\pi/2} \cos^{n-2} x \sin^2 x \, dx$$

②

$$= (n-1) \int_0^{\pi/2} \cos^{n-2} x \{1 - \cos^2 x\} \, dx$$

$$= (n-1) \int_0^{\pi/2} \cos^{n-2} x \, dx - (n-1) \int_0^{\pi/2} \cos^n x \, dx$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$\underline{\underline{I_n = \frac{(n-1)}{n} I_{n-2}}}$$

3 Marks: correct answer

2 Marks: significant progress
up to and including ②
and beyond

1 Mark: correctly applies IBP
to obtain ① and makes
additional relevant progress.

$$ii) \quad \frac{I_n}{I_{n-2}} = \frac{n-1}{n}$$

$$\Rightarrow \frac{I_{2n}}{I_0} = \frac{I_{2n}}{1} \times \frac{1}{I_0}$$

$$= \frac{I_{2n}}{I_{2n-2}} \times \frac{I_{2n-2}}{I_{2n-4}} \times \frac{I_{2n-4}}{I_{2n-6}} \times \dots \times \frac{I_4}{I_2} \times \frac{I_2}{I_0}$$

$$= \frac{2n-1}{2n} \times \frac{2n-3}{2n-2} \times \frac{2n-5}{2n-4} \times \dots \times \frac{3}{4} \times \frac{1}{2} \quad (1)$$

$$= \left(\frac{2n}{2n} \right) \times \left(\frac{2n-1}{2n} \right) \times \left(\frac{2n-2}{2n-2} \right) \times \left(\frac{2n-3}{2n-2} \right) \times \left(\frac{2n-4}{2n-4} \right) \times \left(\frac{2n-5}{2n-4} \right) \times \dots \times \left(\frac{2}{2} \right) \times \left(\frac{1}{2} \right)$$

$$= 2n \times (2n-1) \times (2n-2) \times (2n-3) \times \dots \times 3 \times 2 \times 1 \quad (2)$$

$$(2n)(2n-1)(2n-2)(2n-3) \dots 2(2) \cdot 2(2) \cdot 2(1) \dots$$

$$= \frac{(2n)!}{\dots}$$

$$\underbrace{\{2 \cdot 2 \cdot 2 \dots 2\}}_{2n \text{ times}} \cdot \{n^2 \cdot (n-1)^2 \cdot (n-2)^2 \dots 2^2 \cdot 1^2\}$$

$$= \frac{(2n)!}{2^{2n} \{n(n-1)(n-2) \dots 2 \cdot 1\}^2}$$

$$\frac{I_{2n}}{I_0} = \frac{(2n)!}{2^{2n} (n!)^2}$$

3 Marks : correct answer

1 Mark: Obtains expression (1)

2 Marks: obtains a correct

or equivalent progress

version of expression (2)

5.

$$\text{iii)} \quad \int_0^a f(x) dx = \int_0^{a/2} f(x) dx + \int_{a/2}^a f(x) dx$$

$$\text{Let } u = a - x \Rightarrow x = a - u$$

$$\therefore dx = -du \quad x=0 \Rightarrow u=a, \quad x=\frac{a}{2} \Rightarrow u=\frac{a}{2}, \quad x=a \Rightarrow u=0$$

$$\int_{a/2}^a f(x) dx = \int_{a/2}^0 f(a-u) (-du) \quad (1)$$

$$= - \int_{a/2}^0 f(a-u) du$$

$$= \int_0^{a/2} f(a-u) du$$

$$= \int_0^{a/2} f(a-x) dx$$

$$\therefore \int_0^a f(x) dx = \int_0^{a/2} f(x) dx + \int_0^{a/2} f(a-x) dx$$

$$= \int_0^{a/2} \{ f(x) + f(a-x) \} dx$$

2 Marks : correct answer

1 Mark : makes significant progress beyond (1)

$$\begin{aligned}
 \text{(iv)} \quad \int_0^{\pi} x \cos^6 x \, dx &= \int_0^{\pi/2} \left\{ x \cos^6 x + (\pi - x) \cos^6 (\pi - x) \right\} dx \\
 &= \int_0^{\pi/2} \left\{ x \cos^6 x + \pi \cos^6 (\pi - x) - x \cos^6 (\pi - x) \right\} dx \\
 \cos(\pi - x) &= \cos \pi \cos x + \sin \pi \sin x \\
 &= -1 \cos x + 0 \sin x \\
 &= -\cos x \\
 &= \int_0^{\pi/2} \left\{ x \cos^6 x + \pi \cos^6 (\pi - x) - x \cos^6 x \right\} dx \\
 &= \pi \int_0^{\pi/2} \cos^6 x \, dx
 \end{aligned}$$

$$\begin{aligned}
 I_n &= \left(\frac{n-1}{n} \right) I_{n-2} \Rightarrow I_6 = \pi \left[\frac{5}{6} I_4 \right] \\
 &= \pi \left[\frac{5}{6} \right] \left[\frac{3}{4} I_2 \right] \\
 &= \pi \left[\frac{5}{6} \right] \left[\frac{3}{4} \right] \left[\frac{1}{2} I_0 \right]
 \end{aligned}$$

$$I_0 = \int_0^{\pi/2} (\cos x)^0 \, dx = \int_0^{\pi/2} 1 \, dx = \left[x \right]_0^{\pi/2} = \pi/2$$

$$\int_0^{\pi} x \cos^6 x \, dx = \pi \left(\frac{5}{6} \right) \left(\frac{3}{4} \right) \left(\frac{1}{2} \right) \cdot \pi/2 = \frac{5\pi^2}{32}$$

3 Marks: correct answer
from specified method

2 Marks: Obtains 1 mark AND
determines the sequence
 $\frac{5}{6}, \frac{3}{4}, \frac{1}{2}$

1 Mark: evaluates I_0 or
obtains $\pi \int_0^{\pi/2} \cos^6 x \, dx$